



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
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Seat No.						
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Institute: (0203)SYMBIOSIS CENTRE FOR MANAGEMENT AND HUMAN RESOURCE DEVELOPMENT, PUNE
Programme: (020344) MASTER OF BUSINESS ADMINISTRATION (INFRASTRUCTURE DEVELOPMENT AND MANAGEMENT)
Batch: 2024-26
Semester: I
Course: Business Statistics
Course Code: 0203440102
Date: 12/03/2025
Day: Wednesday
Maximum Marks: 40
Time: 09:30 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instruction:

- 1) All questions are compulsory.

Q.1 A company produces light bulbs, and the lifespan of these bulbs follows a normal distribution with a mean of 1,200 hours and a standard deviation of 100 hours

- What is the probability that a randomly selected light bulb lasts more than 1,350 hours?
- What is the probability that a randomly selected light bulb lasts less than 1,350 hours?
- What is the probability that a randomly selected light bulb lasts between 1150 to 1350?

Explain Normal Disritbution

(CO 1) (10 Marks)

Q.2 A nutritionist claims that the average daily sodium intake of adults in a certain city is 2,500 mg. To test this claim, a random sample of 30 adults is selected, and their average daily sodium intake is found to be 2,650 mg with a standard deviation of 400 mg.

At a 5% significance level, can you conclude that the average daily sodium intake is different from 2,500 mg? Also write a short note on one sample t test

(CO 2) (10 Marks)

Q.3 An HR manager wants to assess whether there is a difference in job satisfaction levels among employees from thrèe different departments: Sales, Marketing, and IT. The manager conducts a survey where employees rate their job satisfaction on a scale from 1 to 100. Fifteen employees are randomly selected, five from each department, and their satisfaction scores are recorded below:

- Sales Department: 72, 75, 78, 74, 77
- Marketing Department: 85, 88, 82, 87, 83
- IT Department: 65, 70, 68, 72, 69

At a 5% significance level, can you conclude that job satisfaction differs between the three departments?

create Null and alternate hypothesis, and give strategies for the same. Also write a short note on ANOVA

(CO 3) (10 Marks)

- Q.4** An HR manager wants to determine if there is an association between the department an employee belongs to and their promotion status. The following data is collected from 120 employees regarding their department and whether they were promoted:

	Promoted	Not Promoted	Total
Sales Department	25	15	40
Marketing Department	20	20	40
IT Department	15	25	40
Total	60	60	120

At a 5% significance level, can you conclude that the promotion status is associated with the department an employee works in?

(CO 4) (10 Marks)
